

TRUMP $\mathcal{E}$: Task-Responsive Unified Multi-modal Perception

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Students: Shih-wen Liu



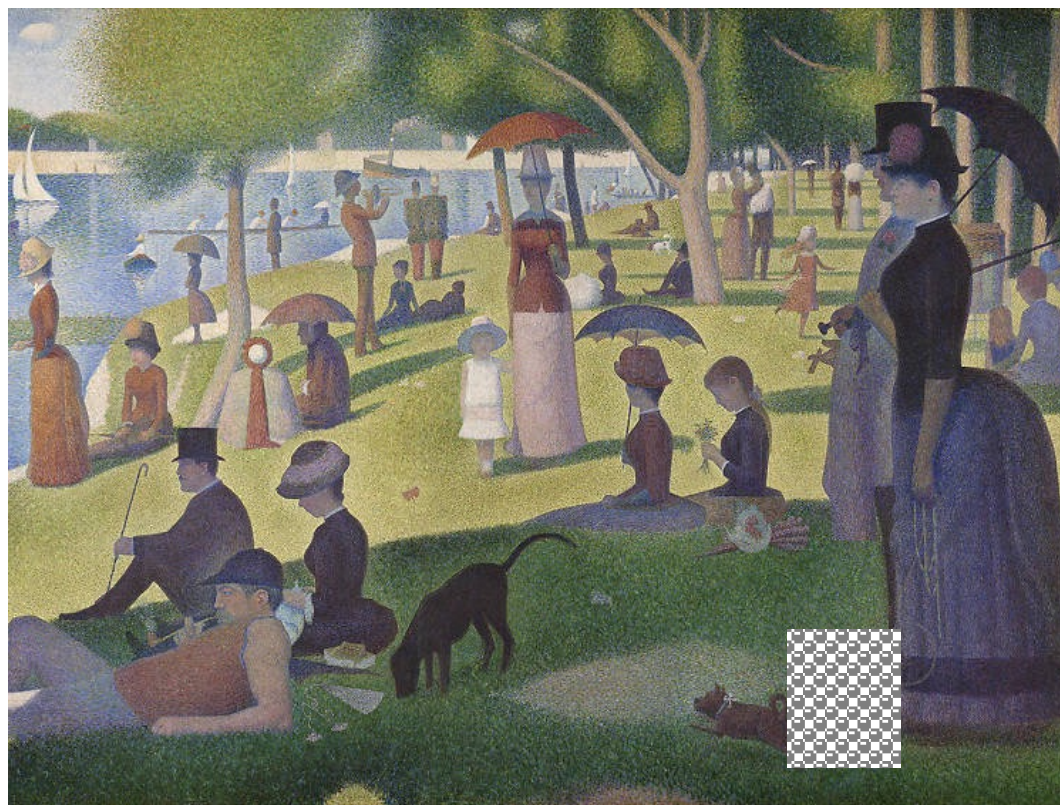
Reasons for this being CVPR 2026 Oral...

- Good Idea, simple concept
- Knowledge transfer
- Plug and play



Georges Seurat, A Sunday Afternoon on the Island of La Grande Jatte, 1884–1886

Question: 右邊女人手裡牽繩的是什麼動物？



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Redundant information

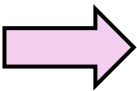
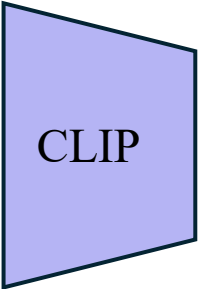
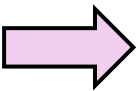
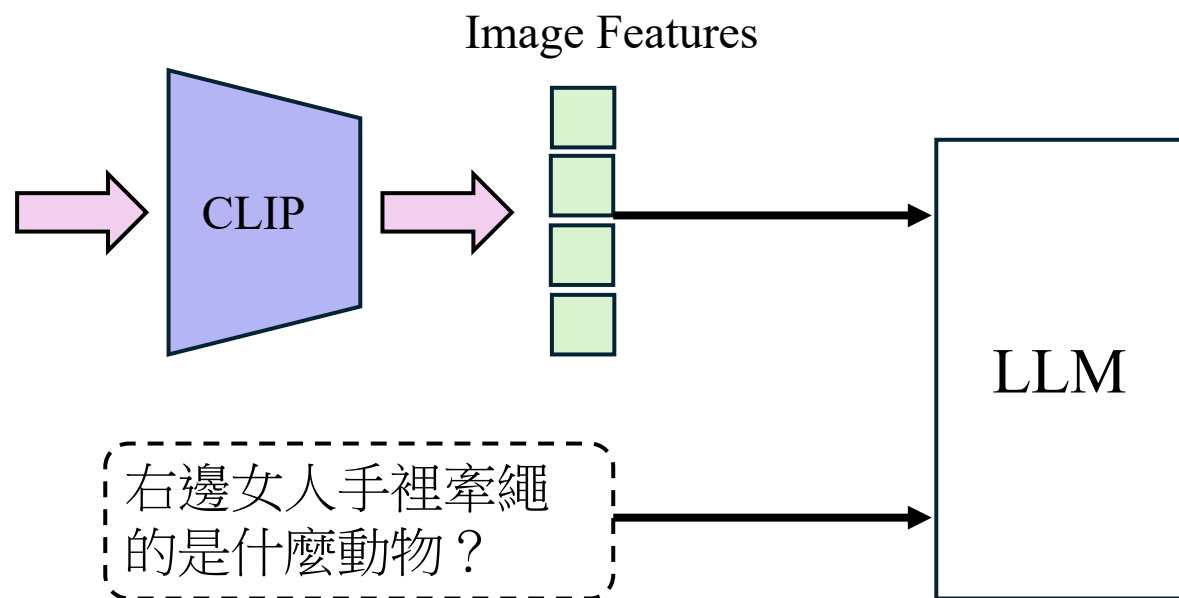


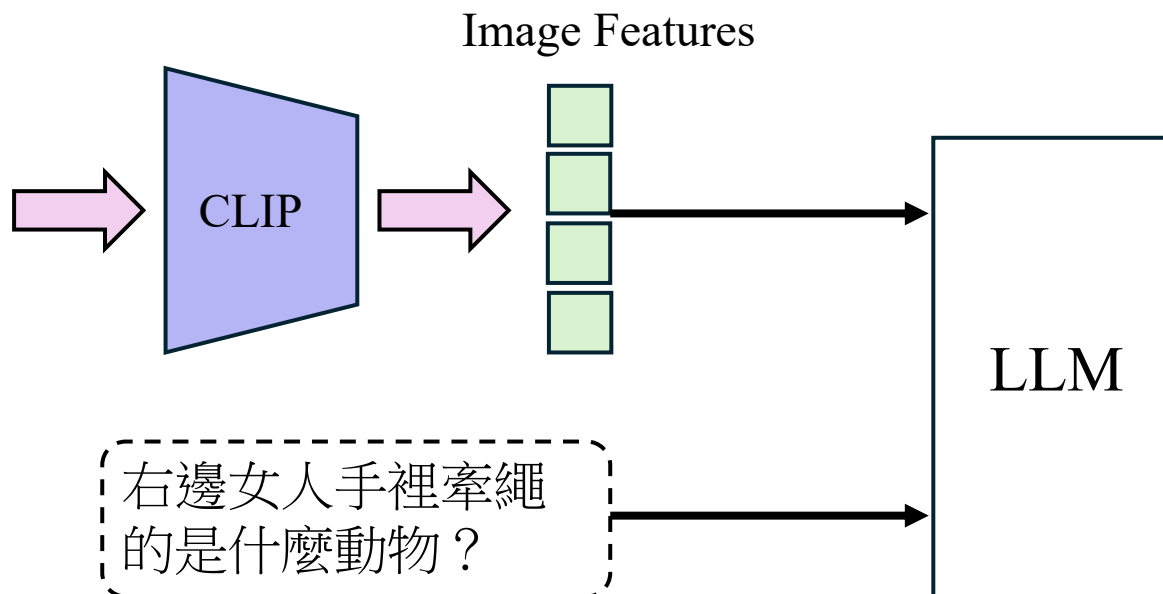
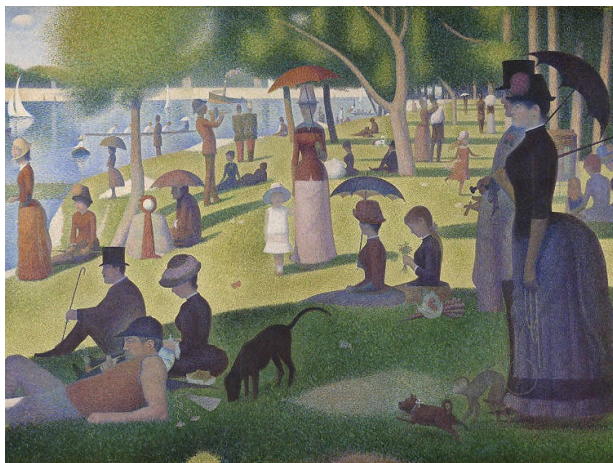
Image Features



Redundant information



Redundant information

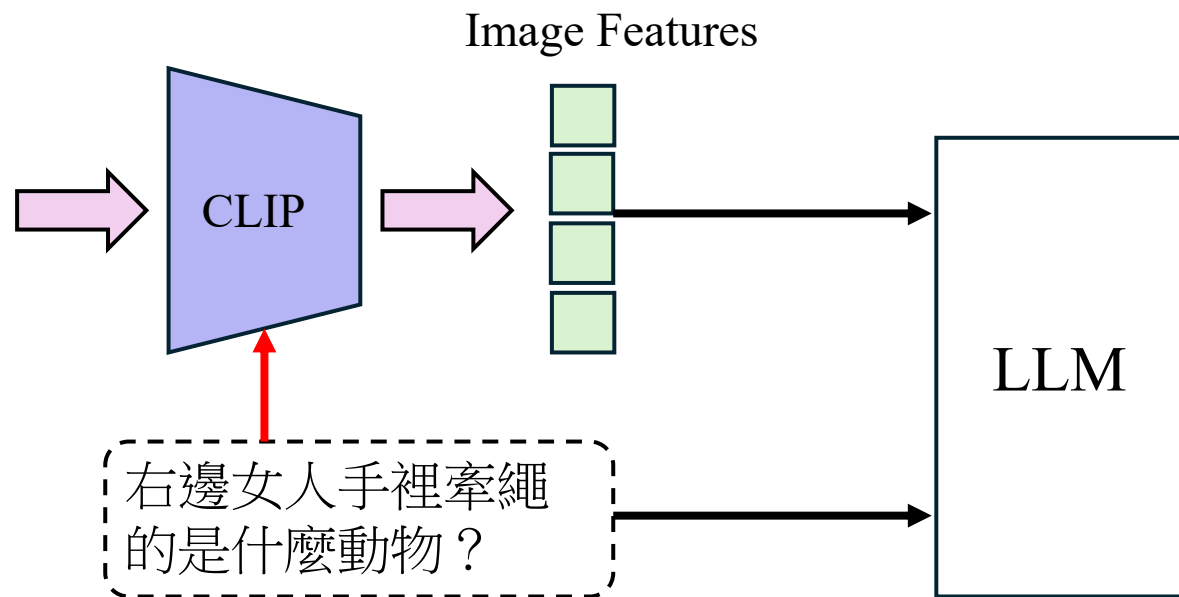


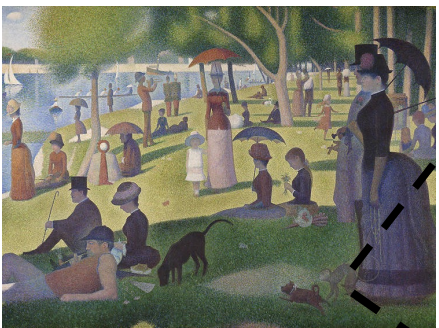
Question: 右邊女人手裡牽繩的是什麼動物？



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Redundant Information





- 風格流派



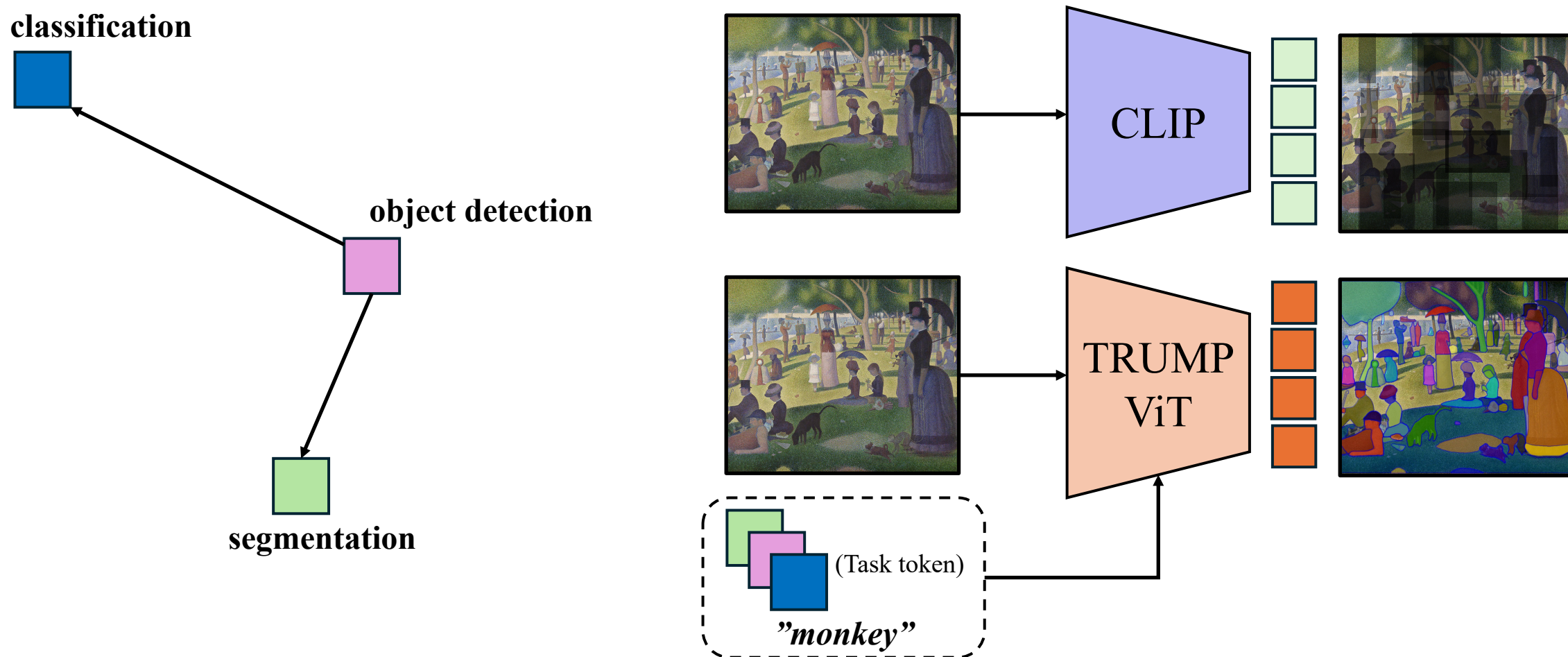
- 猴子品種



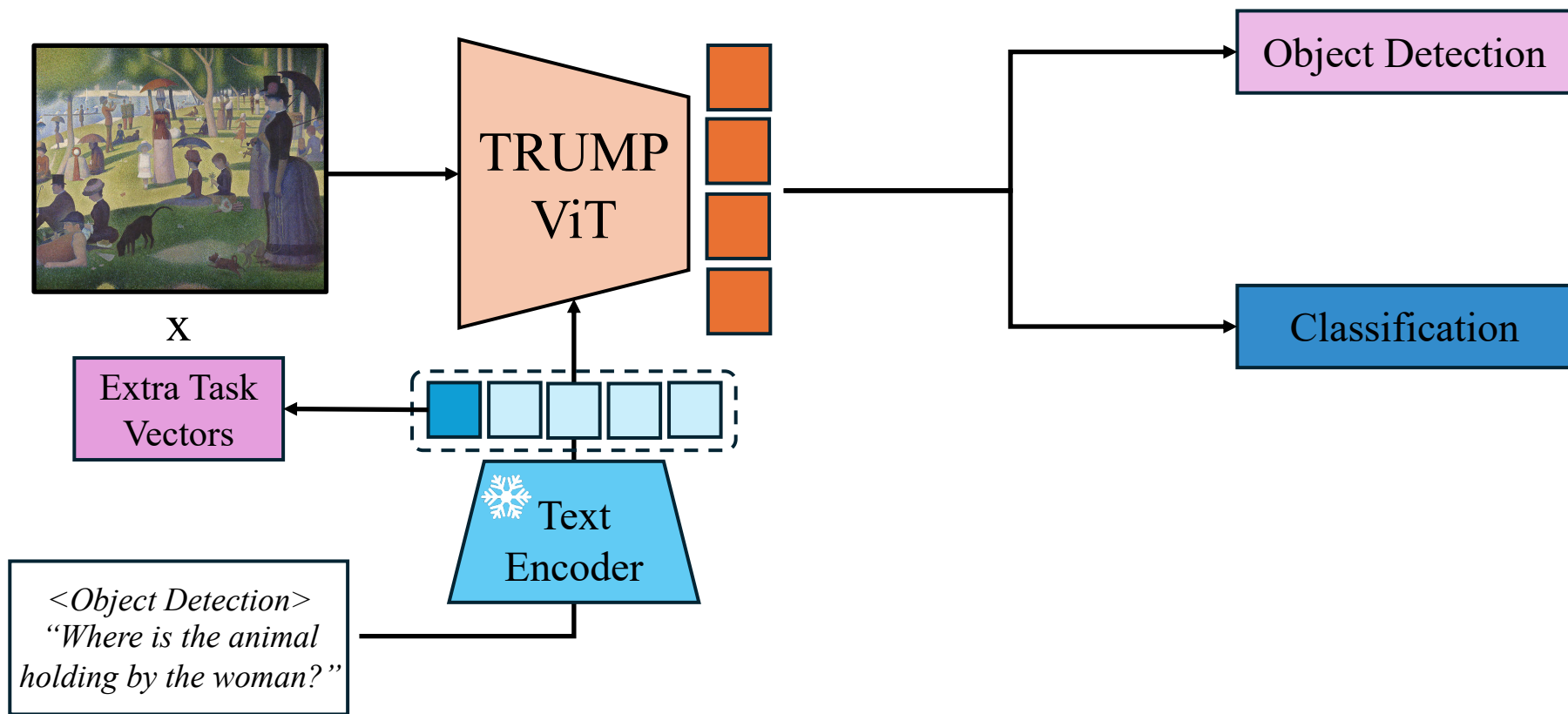
- 養猴子的風氣

Georges Seurat, A Sunday Afternoon on the Island of La Grande Jatte, 1884–1886

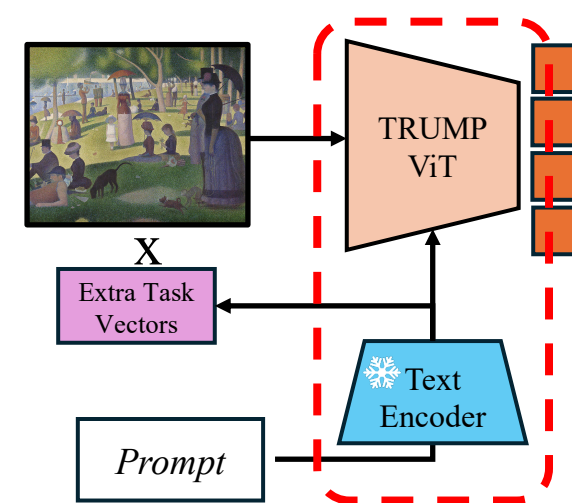
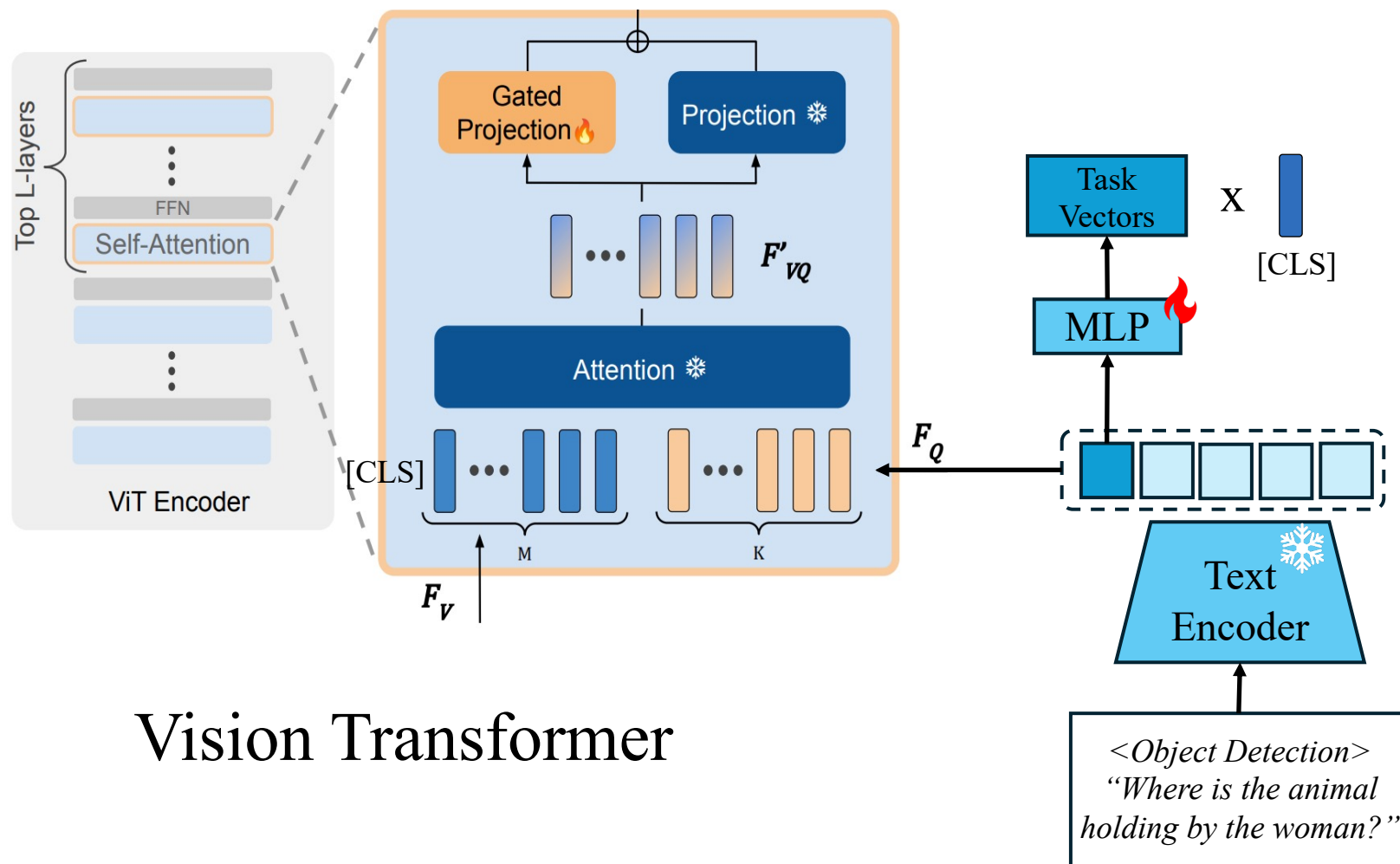
TRUMP: Task-Responsive Unified Multi-modal Perception (framework)



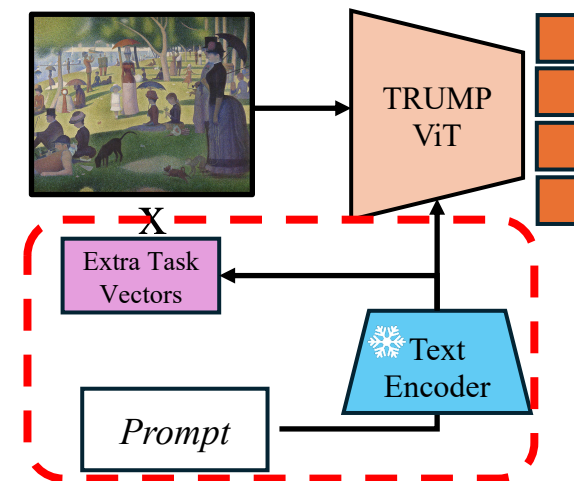
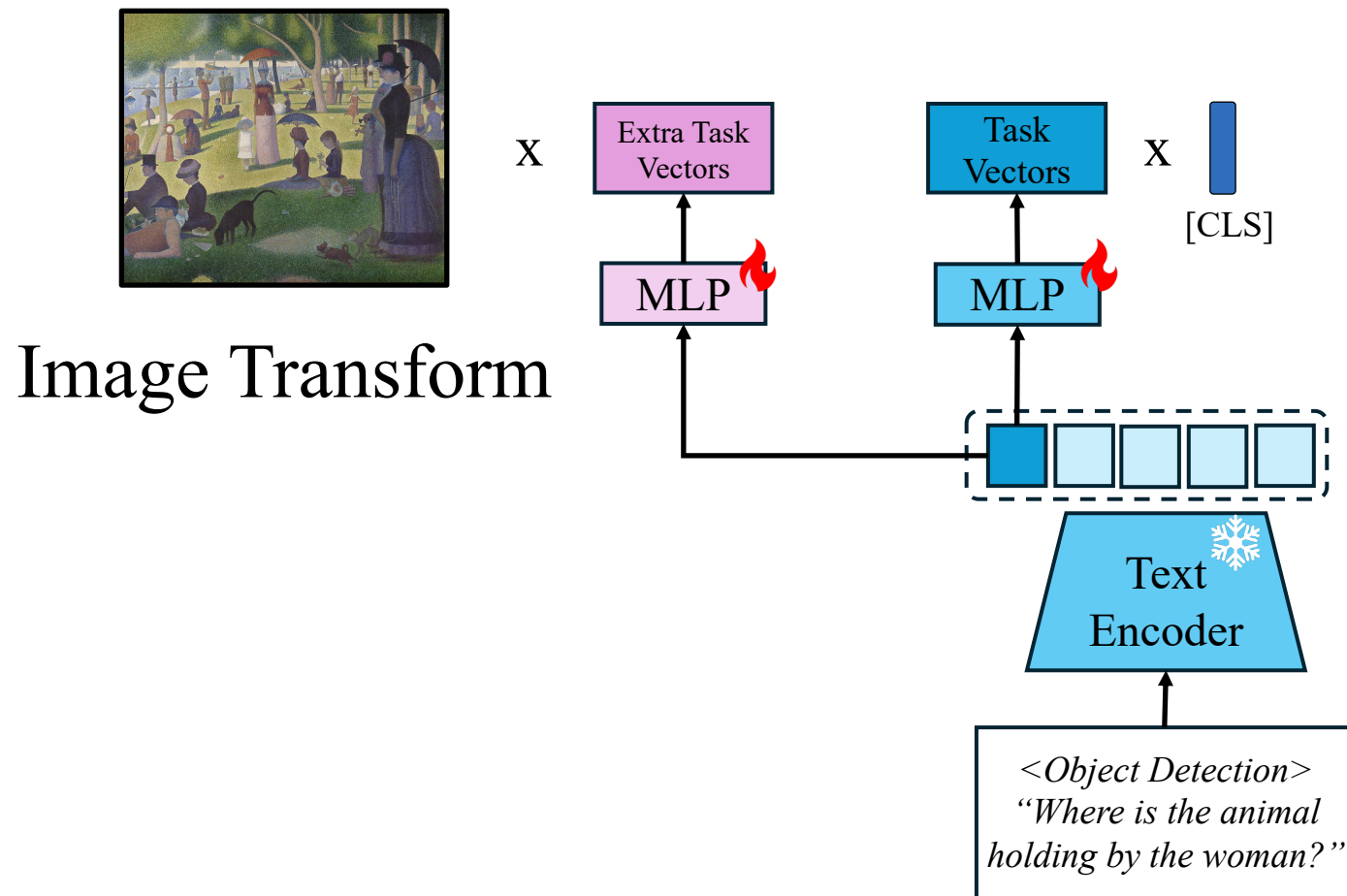
TRUMP: Current Method



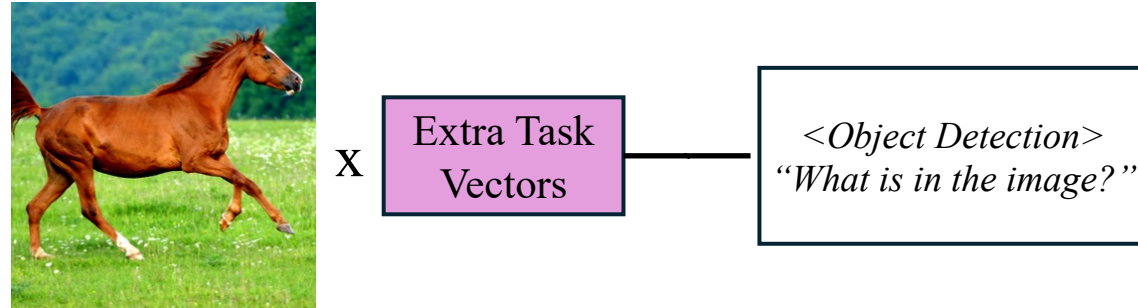
Task Guidance (TG)



Task Vectors (TV)



Task Vectors



(a) Original Horse Image



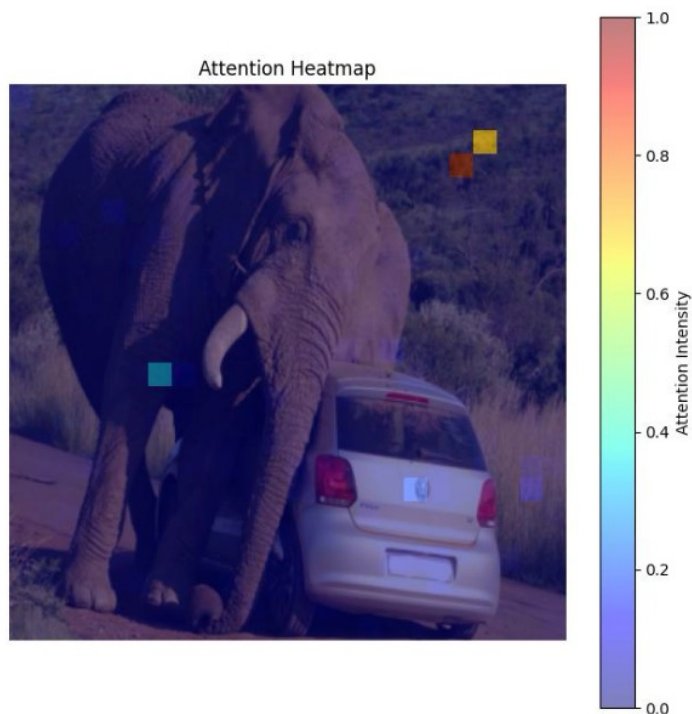
(b) Classification Task Vectors



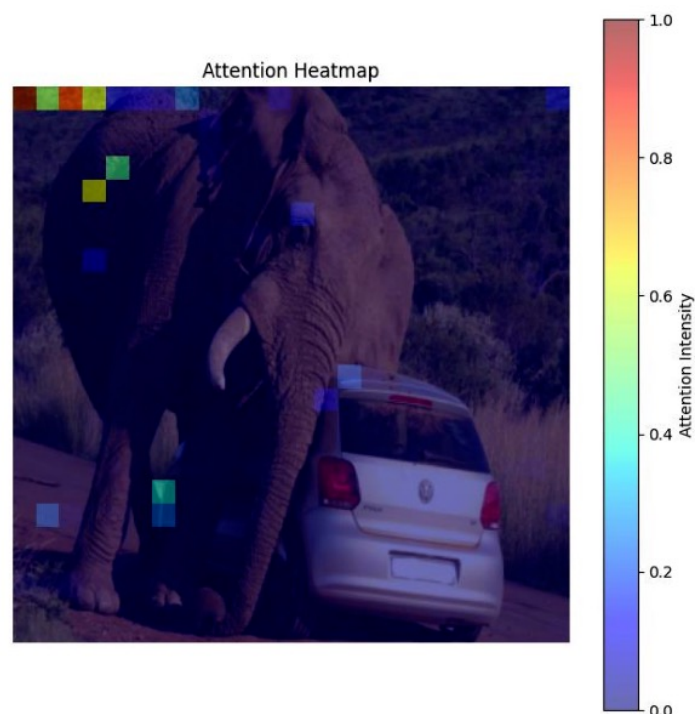
(c) Object Detection Task Vectors

Task Guidance Visualization

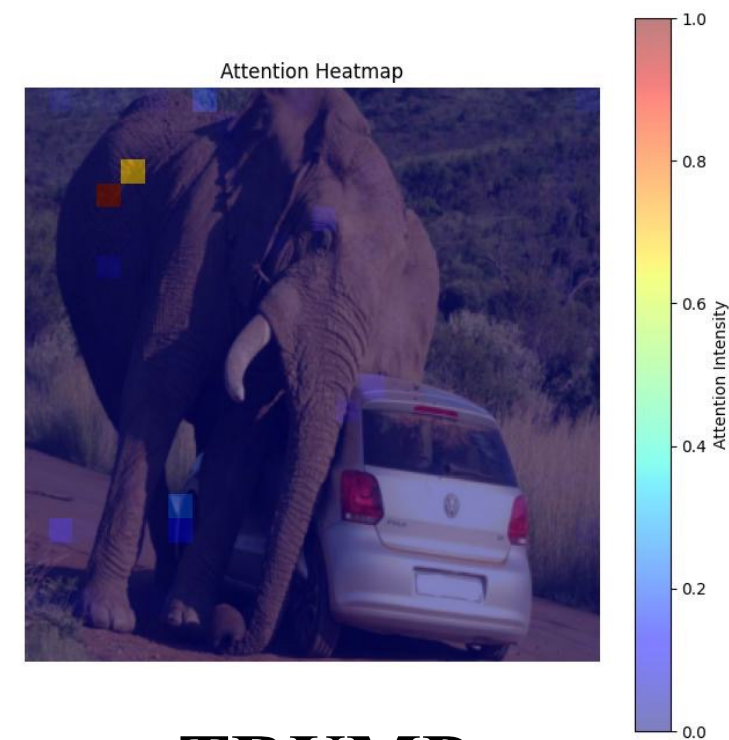
<Object Detection>
“Where is the elephant?”



CLIP-ViT



MTL w/ TG



TRUMP

Experiment Result

TV: Task Vector

TG: Task Guidance

Model	Dataset	Task	Accuracy	mIoU
ClassViT	CIFAR10	Classification	<u>0.9137</u>	-
MTL	CIFAR10	Classification	0.9026	-
MTL w/ TG	CIFAR10	Classification	0.8606	-
MTL w/ TV	CIFAR10	Classification	0.9012	-
TRUMP	CIFAR10	Classification	0.9322	-

Experiment Result

TV: Task Vector

TG: Task Guidance

Model	Dataset	Task	Accuracy	mIoU
ObjectViT	Pascal	Object Detection	0.7192	0.6262
MTL	Pascal	Object Detection	<u>0.7250</u>	0.6337
MTL w/ TG	Pascal	Object Detection	0.6894	0.6286
MTL w/ TV	Pascal	Object Detection	0.6923	0.6227
TRUMP	Pascal	Object Detection	0.7268	<u>0.6263</u>

Future work

